CIS-445 MACHINE LEARNING
Group Homework #3

Write the Group # and only the names of your group members who actively participated in this assignment:
Group #5 – Dylan Shea, Channel Caldwell, Clay Peterson

Logistics

1. Get in touch with your group (see the Groups folder on Blackboard).
2. Discuss and work through all 3 problems collectively with your group via email, the Discussion Forum, Blackboard Collaborate Ultra, and/or MS Teams. (Do not divide the work among group members. Collaborating on all problems may help you perform better on the tests.)
3. Choose one recorder to prepare the final copy (one submission per group) and submit it via the Blackboard Assignments/Homeworks folder by the due date. You must provide your answers on these two sheets.
4. Do **NOT** include the names of group members who did not participate in the assignment or whose contribution was minimal. If you do so and this is discovered, it will be considered academic dishonesty, and **ALL** group members will receive a **0** for this assignment.

Worth 60 points. Due date: See Blackboard

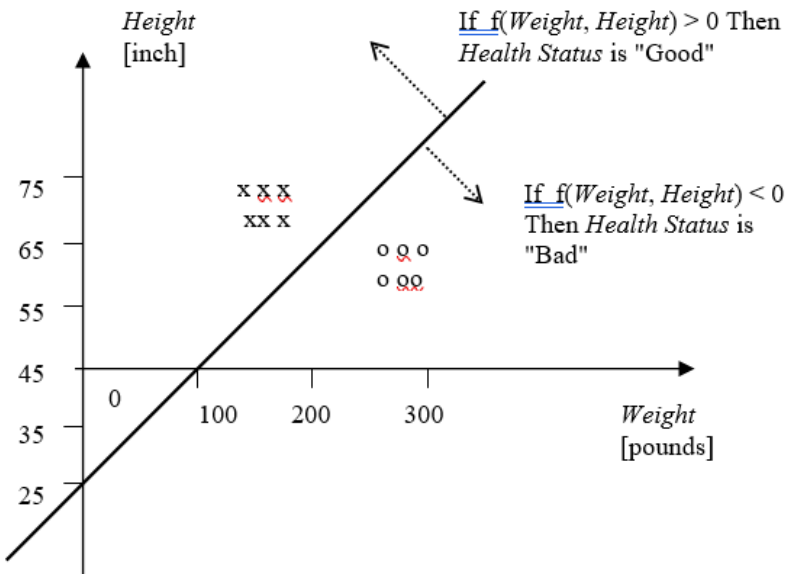
Objectives:

Learn about Classification, Regression, and Clustering. Interpretation of the Results: Confusion Matrix, Correct and Incorrect Classification Rates, MSE, RMSE, R^2, Goodness of Fit, Interpretation of Clusters

(You **MUST** show your calculations!)

Problem 1.

The graph below shows 12 data points denoted by the symbols "x" and "o". These points represent the values for the variables *Weight*, *Height*, and *Health Status* that were used to train a classifier that can separate the data into two classes/patterns: "Good" (x) and "Bad" (o). Based on the 12 data points, a linear function specifying the decision boundary was found. The function (the line) is shown on the graph. The equation of the function is given on page 2. Apply the equation of the function to find the predicted *Health Status* for the following 12 test cases (see table below) that the classifier has not seen during training. Then fill in each entry in column 5 of the table with the appropriate symbols "x" or "o".)



Case #	*Weight*	*Height*	True *Health Status*	Predicted *Health Status*
1	120	65	x	x
2	150	75	x	x
3	140	70	x	x
4	200	60	x	o
5	220	63	x	o
6	160	68	x	x
7	230	60	o	o
8	225	62	o	o
9	240	58	o	o
10	250	61	o	o
11	180	75	o	x
12	200	55	o	o

Evaluate the performance (classification accuracy rates) of the classifier by answering the following questions. Note that in order to perform the evaluation, the True/Actual *Health Status* for each test case (column 4 in the table) has to be known.

See how to find the equation of the function *Height*=f(*Weight*)

The *x* and *y* axes represent *Weight* and *Height*, respectively.

The line (function *Height* =f(*Weight*) crosses the following two points of the coordinates:

(*Weight1*, *Height1*) = (100, 45) and (*Weight2*, *Height2*) = (0, 25).

<u>There is only one line crossing the above 2 points</u>.

Plug in the values for the coordinates to the following equation
From the equation below, one can develop the equation of the function.

$$\frac{Weight - Weigth1}{Weight2 - Weight1} = \frac{Height - Height1}{Height2 - Height1}$$

$$\frac{Weight - 100}{0-100} = \frac{Height - 45}{25 - 45}$$

From the above, express *Height* as a function of *Weight*.

(Weight-100)(-20)=(-100)(Height-45)
(Weight-100)=5(Height-45)
Weight-100=5Height-225
5Height-225=Weight-100
5Height=Weight+125
Height=25 +0.2Weight

For help see pages 19-20 in file *4 - Machine Learning & Data Mining Tasks and Tools.pdf*.

Equation of the function, *Height*=f(*Weight*): **Height = 25 + 0.2 Weight**

To determine the predicted Health Status as ("Good" – x or "Bad" - o) for each of the 12 test cases in the table, you need to plug in the values for Height and Weight for each case and examine the sign.

If (Height – 25 - 0.2Weight) >= 0 Then Health Status = "Good" – x
If (Height – 25 - 0.2Weight) < 0 Then Health Status = "Bad" – o

Case 1: If (65 – 25 - 0.2*120 = 40 – 24 = 16) >= 0 Health Status = "Good" – x
Case 2: If (75– 25 - 0.2*150 = 50 – 30 = 20) >= 0 Health Status = "Good" – x
Case 3: If (70– 25 - 0.2*140= 45 – 28 = 17) >= 0 Health Status = "Good" – x
Case 4: If (60– 25 - 0.2*200 = 35 – 40 = -5) < 0 Health Status = "Bad" – o
Case 5: If (63– 25 - 0.2*220 = 38 – 44 = -6) < 0 Health Status = "Bad" – o
Case 6: If (68– 25 - 0.2*160 = 43 – 32 = 11) >= 0 Health Status = "Good" – x
Case 7: If (60– 25 - 0.2*230 = 35 – 46 = -11) < 0 Health Status = "Bad" – o
Case 8: If (62– 25 - 0.2*225 = 37 – 45 = -8) < 0 Health Status = "Bad" – o
Case 9: If (58– 25 - 0.2*240 = 33 – 48 = -15) < 0 Health Status = "Bad" – o
Case 10: If (61– 25 - 0.2*250 = 36 – 50 = -14) < 0 Health Status = "Bad" – o
Case 11: If (75– 25 - 0.2*180 = 50 – 36 = 14) >= 0 Health Status = "Good" – x
Case 12: If (55– 25 - 0.2*200 = 30 – 40 = -10) < 0 Health Status = "Bad" – o

Calculate the predicted Health Status for Cases 2-12. See case 1 as an example. When you are done, fill in column #5 with the symbol/value (**x** or **o**) for the predicted Health Status.
Create the confusion matrix below to summarize and present the classification results [numbers and percentages]. Label all entries in the matrix appropriately.

True (actual) class		Predicted class	
		Health status good (x)	Health status bad (o)
	Health status good (x)	4 (33.33%)	2 (16.67%)
	Health status bad (o)	1 (8.33%)	5 (41.67%)

What is the number and percentage of the correct classification accuracy rate of good cases?
_____4_____ _____4/12=33.33%_____

What is the number and percentage of the correct classification accuracy rate of bad cases?
_____5_____ _____5/12=41.67%_____

What is the number and percentage of the incorrect classification accuracy rate of good cases?
_____2_____ _____2/12=16.67%_____

What is the number and percentage of the incorrect classification accuracy rate of bad cases?
_____1_____ _____1/12=8.33%_____

What is the number and percentage of the overall correct classification accuracy rate?
_____9_____ ____9/12=75%_____

What is the number and percentage of the overall error rate?
_____3_____ _____3/12=25%_____

What is the number and percentage of the false positive misclassification errors? (tool generated healthy (x) but it should have generated not healthy (bad o))
_____1_____ ____1/12=8.33%_____

What is the number and percentage of the false negative misclassification errors? (tool generated not healthy (bad o) but it should have generated healthy (x))

_____2_____ _____2/12=16.67 **More dangerous** _____

Problem 2

Assume that you ran multiple regression analysis on a historical data set that describes real estate properties. A data set has the following three input attributes (variables): the size of the house [square foot], number of bathrooms, and location of a property; and the output attribute: the sale price [$].

Assume that the regression analysis generated the following coefficients (parameters): 100, 3000, and 10000 for the nonlinear function *price=f(x,y,z)* shown below.

$price = 100x + 3000(y-1)^2 + 10000z$

where variables *x*, *y*, and *z* represent the size of the house, number of bathrooms, and location of a real estate, respectively.

<u>Note</u>

(1) Statisticians often call input variables and an output variable as independent variables and a dependent variable, respectively. (SAS calls an output variable a target variable.)
(2) For the purpose of this exercise I made up the values of 100, 3000, and 10000 for the coefficients. If you had had real data set and run the regression analysis using, for example SAS, the values for the three coefficients would have been different.
(3) In real world applications the number of input attributes used to determine the price of a real estate property would obviously be greater than 3.
(4) In multiple regression analysis knowledge (the concept learned) is encoded in the coefficients standing by the variables *x*, *y*, and *z*. This knowledge is relatively easy to interpret. Other data mining tools such as neural networks encode knowledge in weights (small numerical values connecting neurons) which are often difficult to interpret, whereas decision trees encode knowledge in simple to understand if-then rules.

Using the equation calculate the predicted price of the following real estate properties for the following values of *x*, *y*, and *z*

x	*y*	*z*	true/actual *price* [$]	predicted *price* [$]
1000	3	1	120000	**122,000**
1500	2	3	180000	**183,000**
2000	3	7	281000	**282,000**
6000	5	8	730000	**728,000**
3500	2	5	397000	**403,000**

Calculate the mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and goodness of fit (also called R^2 or coefficient of determination) using the formulas discussed in class.

Show your calculations and discuss the results on the next page. For help see pages 20-21 in file *4 - Machine Learning & Data Mining Tasks and Tools.pdf*.

MSE: **10,800,000** RMSE: 3,286
MAE: 2,800 Goodness of fit or coefficient of determination (R^2): .9998

Interpret the values of RMSE and MAE (low/high?), and R^2 (good/bad and why?) How would you evaluate the predicted results (sale price) produced by the tool (good/bad and why?)? Discuss briefly.

Need to Calculate Error:

F(1000) = 122,000 – 120,000 = 2,000
F(1500) = 183,000 – 180,000 = 3,000
F(2000) = 282,000 – 281,000 = 1,000
F(6000) = 728,000 – 730,000 = -2,000 Absolute Error: 2,000
F(3500) = 403,000 – 397,000 = 6,000

Squared Errors = {
4,000,000 9,000,000 1,000,000 4,000,000 36,000,000}

SSE(Sum of Squared Errors) = 54,000,000

MSE = 54,000,000/5 = 10,800,000

RMSE = 3286.33

MAE = 14,000/5 = 2800

Sum of True Prices = 1,708,000, Mean of True = 341,600

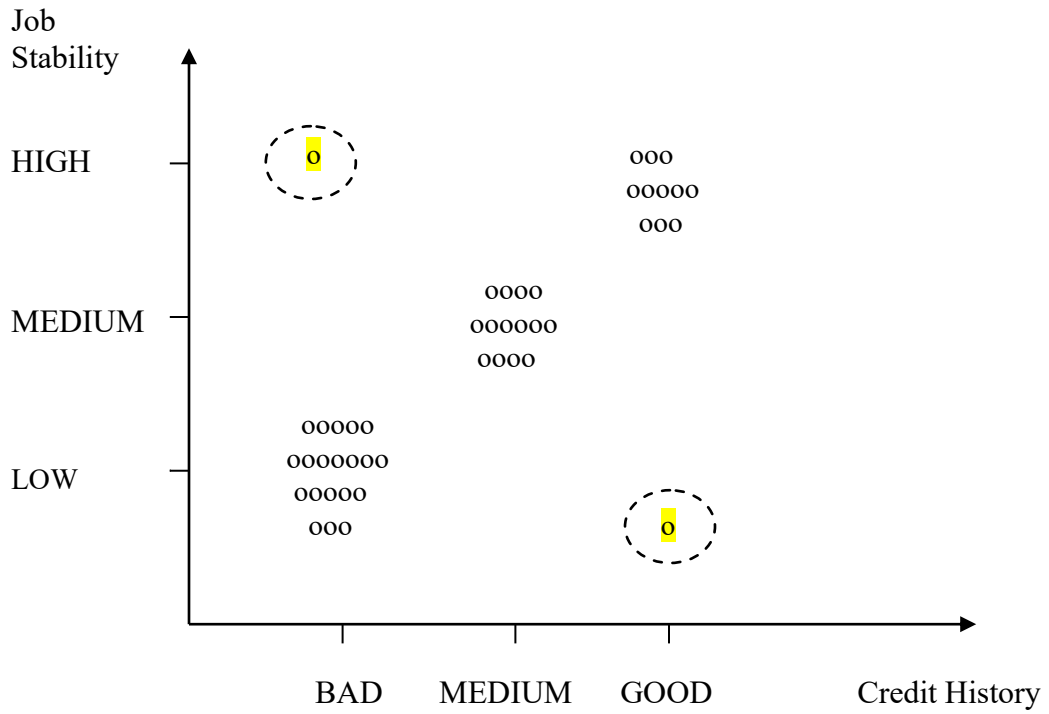
True Price	TP - True Mean	TP*TM ^ 2
120000	-221600	49106560000
180000	-161600	26114560000
281000	-60600	3672360000
730000	388400	1.50855E+11
397,000	55400	3069160000
		2.32817E+11

SST = 2.3817E+11 = 238,170,000,000

SSE/SST = 0.00023, R^2 = 1 – 0.00023 = .9998

RMSE and MAE are both extremely low, which is due to the low errors within the pricing. R^2 is a near perfect fit, as there is little to no error. The model overall is extremely accurate

Problem 3: A bank loan officer retrieved from the data warehouse 47 records representing 2 variables of customers of the bank and plotted them on a 2-dimensional graph below. To represent the data values a symbol "o" is used. The *x* and *y* variables represent the Credit History and Job Stability of a customer, respectively. Both variables *x* and *y* are expressed using the linguistic terms such as BAD, MEDIUM, GOOD, and LOW, MEDIUM, HIGH, respectively.



Note: This problem is about clustering or data segmentation that uses unsupervised learning. In this type of learning there is no output variable with predefined classes/outcomes.

How many clusters can the bank loan officer distinguish in the data? __**3 Clusters (Bad, Low), (Medium, Medium), (Good, High)**_____. Denote these clusters.

How might he/she interpret these clusters? (See the similar example on page 16 in file *4 - Machine Learning & Data Mining Tasks and Tools.pdf*.

The first cluster, (Bad,Low) can be interpreted as a high risk customer because of their bad credit and low job stability.

The Second cluster, (Medium,Medium) can be interpreted as a medium risk customer because they have a medium credit score and their job stability is medium as well.

The third cluster, (Good, High), can be interpreted as a low risk customer because they have a high credit score and has high job stability.

Could he/she perhaps assign a numeric value to each cluster to represent a particular class and use these classes for supervised learning?
Yes, you can give numeric values to each cluster to represent a particular class, you can give the (Bad, Low) cluster the numerica value of 0, the (Medium, Medium) cluster 1, and the (Good, High) cluster 2. These classes can be used in supervised learning because if a customer has a low score the model can flag that customer for credit risks.

Circle the outliers, if any, on the graph.